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United States Patent Application Publication	20250280190
Kind Code	A1
Publication Date	September 04, 2025
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IMAGE PICKUP APPARATUS INCLUDING BUILT-IN OPTICAL FILTER, CONTROL METHOD OF IMAGE PICKUP APPARATUS, AND STORAGE MEDIUM

Abstract

An image pickup apparatus includes an image sensor, an optical filter movable between a first position at which the optical filter is inserted into an imaging range of the image sensor and a second position at which the optical filter is retracted from the imaging range, and a control unit configured to perform exposure control for the image sensor in a case where the optical filter is located at the first position and an in-focus state cannot be obtained by moving a focus lens.

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Family ID: **87832816**

Appl. No.: **19/210577**

Filed: **May 16, 2025**

Foreign Application Priority Data

JP	2022-032364	Mar. 03, 2022

Related U.S. Application Data

parent US continuation 18176740 20230301 parent-grant-document US 12335606 child US 19210577

Publication Classification

Int. Cl.: **H04N23/63** (20230101); **H04N23/55** (20230101); **H04N23/667** (20230101); **H04N23/67** (20230101); **H04N23/72** (20230101)

U.S. Cl.:

CPC **H04N23/634** (20230101); **H04N23/55** (20230101); **H04N23/667** (20230101); **H04N23/67** (20230101); **H04N23/72** (20230101);

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] This application is a continuation of application Ser. No. 18/176,740, filed Mar. 1, 2023, the entire disclosure of which is hereby incorporated by reference.

BACKGROUND
Technical Field
[0002] The disclosure relates to an image pickup apparatus having a built-in optical filter, a control method of the image pickup apparatus, and a storage medium.
Description of the Related Art
[0003] An image pickup apparatus including an optical filter such as a light attenuating filter (neutral density (ND) filter) has conventionally been known. In a case where a flange focal length is adjusted in a camera having a built-in ND filter so that an imaging plane is in focus while the ND filter is not used, an optical path length changes after the ND filter is used, and thus the imaging plane may not be focused on a close object. Japanese Patent Laid-Open No. (“JP”) 2002-229116 discloses an image pickup apparatus having a warning unit configured to warn a user that an optical filter is or is not used.
[0004] The image pickup apparatus disclosed in JP 2002-229116 may not be focused on a close object while the optical filter is used.
SUMMARY
[0005] One of the aspects of the embodiment provides an image pickup apparatus that can provide an in-focus state on an object in a case where the in-focus state cannot be obtained by moving a focus lens while an optical filter is used.
[0006] An image pickup apparatus according to one aspect of the disclosure includes an image sensor, an optical filter movable between a first position at which the optical filter is inserted into an imaging range of the image sensor and a second position at which the optical filter is retracted from the imaging range, and at least one processor, and a memory coupled to the at least one processor, the memory having instructions that, when executed by the processor, perform operations as a control unit configured to perform exposure control for the image sensor in a case where the optical filter is located at the first position and an in-focus state cannot be obtained by moving a focus lens. A control method of the above image pickup apparatus and a storage medium storing a program that causes a computer to execute the above control method also constitute another aspect of the disclosure.
[0007] Further features of the disclosure will become apparent from the following description of exemplary embodiments with reference to the attached drawings. In the following, the term “unit” may refer to a software context, a hardware context, or a combination of software and hardware contexts. In the software context, the term “unit” refers to a functionality, an application, a software module, a function, a routine, a set of instructions, or a program that can be executed by a programmable processor such as a microprocessor, a central processing unit (CPU), or a specially designed programmable device or controller. A memory contains instructions or program that,

when executed by the CPU, cause the CPU to perform operations corresponding to units or functions. In the hardware context, the term "unit" refers to a hardware element, a circuit, an assembly, a physical structure, a system, a module, or a subsystem. It may include mechanical, optical, or electrical components, or any combination of them. It may include active (e.g., transistors) or passive (e.g., capacitor) components. It may include semiconductor devices having a substrate and other layers of materials having various concentrations of conductivity. It may include a CPU or a programmable processor that can execute a program stored in a memory to perform specified functions. It may include logic elements (e.g., AND, OR) implemented by transistor circuits or any other switching circuits. In the combination of software and hardware contexts, the term "unit" or "circuit" refers to any combination of the software and hardware contexts as described above. In addition, the term "element," "assembly," "component," or "device" may also refer to "circuit" with or without integration with packaging materials.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0008] FIGS. **1**A and **1**B are external perspective views of an image pickup apparatus according to this embodiment.
[0009] FIG. **2** is a block diagram of the image pickup apparatus according to this embodiment.
[0010] FIG. **3** is an exploded perspective view of the image pickup apparatus according to this embodiment.
[0011] FIGS. **4**A to **4**C explain during driving of an optical filter in this embodiment.
[0012] FIG. **5** is a sectional view of the image pickup apparatus according to this embodiment.
[0013] FIGS. **6**A to **6**C explain an optical path of light incident on an image sensor in this embodiment.
[0014] FIG. **7** illustrates a relationship between a depth of field and an imaging position in this embodiment.
[0015] FIG. **8** explains a defocus map in this embodiment.
[0016] FIG. **9** is a flowchart of autofocus (AF) in a totally automatic imaging mode according to this embodiment.
[0017] FIG. **10** is a flowchart of AF in an F-number priority imaging mode in this embodiment.
[0018] FIG. **11** is a flowchart of AF in a shutter speed priority imaging mode in this embodiment.
[0019] FIG. **12** is a flowchart of AF while the optical filter according to this embodiment maintains a state selected by a user.
DESCRIPTION OF THE EMBODIMENTS
[0020] Referring now to the accompanying drawings, a detailed description will be given of embodiments according to the disclosure. Corresponding elements in respective figures will be designated by the same reference numerals. This embodiment is an example for realizing the disclosure, and should be properly modified or changed according to the configuration of the apparatus to which the disclosure is applied and a variety of conditions, and the disclosure is not limited to the following embodiment. Parts of the embodiment described below may be combined as appropriate.
[0021] Referring now to FIGS. **1**A and **1**B, a description will be given of an image pickup apparatus according to this embodiment. FIGS. **1**A and **1**B are external perspective views of an image pickup apparatus (camera body) **100** according to this embodiment. FIG. **1**A is an external perspective view of the image pickup apparatus **100** viewed from a front side, and illustrates a state in which a lens apparatus (interchangeable lens) **104** (see FIG. **2**) attachable to the image pickup apparatus **100** is detached. FIG. **1**B is an external perspective view of the image pickup apparatus **100** viewed from a rear side. This embodiment describes an imaging system in which the lens

apparatus **104** is attachable to and detachable from the image pickup apparatus (camera body) **100**, but is not limited to this example, and is applicable to an image pickup apparatus in which the camera body and the lens apparatus are integrated with each other.

[0022] The image pickup apparatus **100** includes a grip portion **101** to be gripped by a user to stably hold the image pickup apparatus **100**. A shutter button **102** that is a switch for starting imaging is provided on a top portion of the grip portion **101**. A mount portion (lens mount) **103** is provided in front of the image pickup apparatus **100**, and the lens apparatus **104** is attachable to and detachable from the image pickup apparatus **100** via the mount portion **103**. A mount contact **105** electrically connects the image pickup apparatus **100** and the lens apparatus **104**, supplies power to the lens apparatus **104**, and performs lens control and communication about lens data through an electric signal. In interchanging the lens apparatus **104**, an engagement is released by pressing a lens unlock button **106**, and the lens apparatus **104** can be detached.

[0023] A power switch **107** is used to power on or off the image pickup apparatus **100**. A main electronic dial **108** and a sub electronic dial **119** are rotary operation members that are clockwise and counterclockwise rotatable, and the rotating operation can change a variety of set values such as an F-number and a shutter speed. A mode switching dial **109** is an operation unit for switching an imaging mode among a variety of modes such as a shutter speed priority imaging mode, an F-number priority imaging mode, and a moving (or motion) image capturing mode. A SET button **110** is a push button and is mainly used to determine selection items.

[0024] A liquid crystal monitor **111** displays a variety of setting screens, captured images, and live-view images of the image pickup apparatus **100**. An electronic viewfinder (EVF) **112** is a viewfinder that can be used to display a variety of setting screens, captured images, and live-view images of the image pickup apparatus **100**. A multifunction button **113** is a push button, and the user can arbitrarily assign and use a variety of setting switches relating to imaging. A display panel **114** displays a variety of set states of the image pickup apparatus **100** such as an imaging mode and an ISO speed. The display panel **114** is displayed even if the image pickup apparatus **100** is powered off.

[0025] An accessory shoe **115** has an accessory contact point **116**, and a variety of accessories such as an external strobe and microphone can be attached to it. A media slot lid **173** can be opened and closed, and in a case where it is open, an external recording medium **148** (see FIG. **2**) such as an SD card can be inserted into or ejected from an internal media slot (not illustrated).

[0026] Referring now to FIG. **2**, a description will be given of an electric configuration and operation of the image pickup apparatus **100**. FIG. **2** is a block diagram of the image pickup apparatus **100** and illustrates a state in which the lens apparatus **104** is attached. Elements common to those in FIGS. **1**A and **1**B will be designated by the same reference numerals.

[0027] An MPU **130** is a small central processing unit (CPU) (control unit) built in the image pickup apparatus **100**. A time measuring circuit (TMC) **131**, a shutter driving circuit **132**, a switch sense circuit **133**, a power supply circuit **134**, a battery check circuit **135**, a video signal processing circuit **136**, an optical filter driving circuit **137**, and a piezoelectric element driving circuit **145** are connected to the MPU **130**. The MPU **130** controls the operation of the image pickup apparatus **100**, processes input information, and instructs and controls each element. The MPU **130** has an EEPROM, which can store time information from the time measuring circuit **131** and various setting information.

[0028] The MPU **130** also communicates with a lens control circuit **138** built in the lens apparatus **104** via the mount contact **105**. Thereby, the MPU **130** can control the operations of a focus lens **141** and an electromagnetic diaphragm (diaphragm, aperture stop) **142** via an AF driving circuit **139** or a diaphragm driving circuit **140**. Although FIG. **2** schematically illustrates a single focus lens **141** as an imaging optical system of the lens apparatus **104**, the imaging optical system actually includes many lens units.

[0029] The AF driving circuit **139** is connected, for example, to a stepping motor (not illustrated) and drives the focus lens **141**. The MPU **130** calculates a focus lens driving amount according to a detected defocus amount using the focus signal read out of the image sensor **121** and transmits a focus command including a focus lens driving amount to the lens control circuit **138**. The lens control circuit **138** that has received the focus command controls driving of the focus lens **141** through the AF driving circuit **139**. Thereby, autofocus (AF) is performed.

[0030] The diaphragm driving circuit **140** is connected to a diaphragm actuator such as a stepping motor (not illustrated) and drives a plurality of aperture blades (not illustrated) that form an aperture in the electromagnetic diaphragm (aperture stop) **142**. Driving a plurality of aperture blades can change an aperture size (aperture diameter) and adjust a light amount.

[0031] The MPU **130** calculates a diaphragm driving amount of the electromagnetic diaphragm **142** based on a luminance signal read out of the image sensor **121** and transmits a diaphragm command including the diaphragm driving amount to the lens control circuit **138**. That is, the MPU **130** communicates with the lens control circuit **138** to control the electromagnetic diaphragm **142**. The lens control circuit **138** that has received the diaphragm command controls the driving of the electromagnetic diaphragm **142** through the diaphragm driving circuit **140**. Thereby, a proper aperture value (F-number) is automatically set.

[0032] A mechanical focal plane shutter (FPS) **150** is driven by the shutter driving circuit **132**. During imaging, a front curtain shutter (not illustrated) is moved to open the shutter when the user presses the shutter button **102**, and a rear curtain shutter (not illustrated) is moved according to the desired exposure time to close the shutter. The exposure time to the image sensor **121** is thereby controlled.

[0033] An optical filter **160** is an optical element that imparts a special effect to an image by diffusing incident light or attenuating a specific wavelength range. The optical filter **160** includes an ND filter that attenuates an incident light amount at a constant rate, a polarized light (PL) filter that suppresses reflected light using a polarization film, and a soft filter that diffuses light to create a soft expression, but the optical filter **160** is not limited to these filters. The optical filter **160** can be driven and its position is movable by the optical filter driving circuit **137**. A detailed configuration of the optical filter **160** will be described below.

[0034] An imaging unit **120** mainly includes an optical low-pass filter **122**, an optical low-pass filter holder **123**, a piezoelectric element (piezoelectric member) **124**, and an image sensor **121** each configured as a unit. The image sensor **121** photoelectrically converts an object image (optical image) formed through the lens apparatus **104**. In this embodiment, the image sensor **121** is a Complementary Metal-Oxide-Semiconductor (CMOS) sensor, but is not limited to this example. The image sensor **121** may use a Charge-Coupled Device (CCD) sensor, a Charge Injection Device (CID) sensor, or the like. The optical low-pass filter **122** disposed in front of the image sensor **121** is a single birefringent plate made of crystal and has a rectangular shape. The piezoelectric element **124** is a single-plate piezoelectric element (piezo element), and is vibrated by a piezoelectric element driving circuit **145** that receives an instruction from the MPU **130** and transmits the vibration to the optical low-pass filter **122**. Fine dust attached to the optical low-pass filter **122** can be shaken off by this vibration.

[0035] The video signal processing circuit **136** governs overall image processing such as filtering and data compression processing for the electric signal obtained from the image sensor **121**. Image data for monitor display from the video signal processing circuit **136** is displayed on the liquid crystal monitor **111** and the electronic viewfinder **112** via the liquid crystal driving circuit **144**. The video signal processing circuit **136** can also store image data in a buffer memory **147** through a memory controller **146** according to an instruction from the MPU **130**. The video signal processing circuit **136** can also perform image data compression processing such as JPEG. In a case where continuous imaging is performed, image data can be temporarily stored in the buffer memory **147** and unprocessed image data can be sequentially read out through the memory controller **146**. Thereby, the video signal processing circuit **136** can sequentially perform image processing and

compression processing regardless of the speed of input image data.
[0036] The memory controller **146** has a function of storing image data in the external recording medium **148** and a function of reading image data stored in the external recording medium **148**. The external recording medium **148** is an SD card, a CF card, or the like that is removable from the image pickup apparatus **100**, but is not limited to these examples.
[0037] The switch sense circuit **133** transmits an input signal to the MPU **130** according to the operation state of each switch. A switch SW**1** (**102***a*) is turned on by a first stroke of the shutter button **102** (half-pressing). A switch SW**2** (**102***b*) is turned on by a second stroke of the shutter button **102** (full pressing). The switch SW**2** (**102***b*) when turned on transmits an instruction to start imaging to the MPU **130**. Connected to the switch sense circuit **133** are the main electronic dial **108**, the mode switching dial **109**, the power switch **107**, the SET button **110**, the multifunction button **113**, and the like.
[0038] The MPU **130** communicates information via the accessory communication control circuit **118** the accessory contact **116** to use a function of an unillustrated accessory. The power supply circuit **134** distributes and supplies power from a battery **143** to each element in the image pickup apparatus **100**. The battery check circuit **135** is connected to the battery **143** and notifies the MPU **130** of remaining amount information about the battery **143** and the like.
[0039] Referring now to FIG. **3**, a description will be given of an internal configuration of the image pickup apparatus **100**. FIG. **3** is an exploded perspective view of the image pickup apparatus **100**. The image pickup apparatus **100** has a structure that is mainly covered by an exterior including a front cover **10**, a top cover **11**, and a rear cover **12**, and operation members and display members are attached to each exterior. A holding member **200**, an imaging unit **120**, and a main board (control board) **180** are arranged in order from the object side along an optical axis (OP) **1000** of the imaging optical system, and the optical filter **160** is mounted on the holding member **200**. The optical filter **160** may use any optical member such as an ND filter, a PL filter, a soft filter (low-pass filter), and the like. Moreover, even in a case where the optical filter **160** is not inserted, the image pickup apparatus **100** can be normally operated.
[0040] Referring now to FIGS. **4**A to **4**C, a description will be given of the configuration and switching operation of the optical filter **160**. FIGS. **4**A to **4**C explain the driving of the optical filter **160**, and illustrate the state of the components relating to the optical filter **160** and the holding member **200** viewed from the rear side of the image pickup apparatus **100**.
[0041] The optical filter **160** is inserted into and held by the holding member **200**. The holding member **200** has a gear shape **200***a* and is attached to the front cover **10** so as to be rotatable about the gear shape **200***a*. A motor (actuator, driving unit) **201** for driving the optical filter **160** is attached to the front cover **10**, and a worm gear **201***a* is attached to a driving shaft of the motor **201**. The worm gear **201***a* can rotate the holding member **200** by transmitting a rotational force to the gear shape **200***a* of the holding member **200** via an intermediate gear **202**. A battery housing **170** for accommodating the battery **143** is provided on the right side of the front cover **10**.
[0042] This embodiment assigns a switching function of insertion/retraction of the optical filter **160** to the multifunction button **113**. A description will now be given of the operation for driving the optical filter **160** by the operation of the user. Another button, a dial, a switch, a setting screen, etc. may be used for the switching operation of the optical filter **160**.
[0043] FIG. **4**A illustrates a state (inserted state) in which the optical filter **160** is inserted into an opening **190** provided inside the mount portion **103** of the image pickup apparatus **100** and overlaps the opening **190**. The opening **190** defines the imaging range. Thereby, the light incident on the image sensor **121** passes through the optical filter **160**, and a variety of imaging expressions are available due to the effect of the optical filter **160**. For example, in a case where an ND filter is inserted as the optical filter **160**, incident light is attenuated, and long-exposure imaging and overexposure suppression can be acquired even in a bright environment.
[0044] In a case where the multifunction button **113** is pressed in the inserted state illustrated in

FIG. **4**A, the switch sense circuit **133** detects the pressing. At this time, The MPU **130** transmits an instruction to drive the motor **201**, and the motor **201** starts rotating through the optical filter driving circuit **137**. The rotation of the motor **201** is transmitted from the worm gear **201***a* to the gear shape **200***a* via the gear shape **200***a*, and the holding member **200** having the gear shape **200***a* starts rotating. FIG. **4**B illustrates a state in the middle of the rotation (intermediate state).
[0045] The holding member **200** moves to a retracted state illustrated in FIG. **4**C through the intermediate state illustrated in FIG. **4**B. A rotation shaft of the gear shape **200***a* is located between the opening **190** and the bottom surface side of the image pickup apparatus **100** and between the optical axis **1000** and the short side of the opening **190** on the grip portion **101** side. Thereby, as illustrated in FIG. **4**B, the holding member **200** can be rotated without interfering with internal parts such as the top cover **11**. The motor **201** is also disposed closer to the bottom surface than the optical axis **1000** and the opening **190**, the power transmission distance to the gear shape **200***a* is short, and the driving efficiency is enhanced.
[0046] In a case where the holding member **200** moves to the retracted position (second position) illustrated in FIG. **4**C, the optical filter driving circuit **137** stops the motor **201** in response to a detection signal from a position sensor (not illustrated). The position sensor refers to a position detector such as a photo-reflector, but the driving stop timing may be determined based on any means, such as a detection of the rotation angle of the motor **201** and a mechanical switch.
[0047] FIG. **4**C illustrates a state (retracted state) in which the holding member **200** is rotated by approximately 90 degrees on a plane parallel to the imaging unit **120** from the inserted state illustrated in FIG. **4**A and retracted from the opening **190**. By retracting the holding member **200**, the light condensed by the lens apparatus **104** enters the image sensor **121** without passing through the optical filter **160**.
[0048] FIG. **5** is a sectional view of the image pickup apparatus **100**, illustrating a section taken along a line A-A in FIG. **4**C. As illustrated in FIG. **5**, the holding member **200** is retracted to a space between the opening **190** and the battery housing **170**, that is, between the opening **190** and the grip portion **101**. By rotating the holding member **200** by approximately 90 degrees from the inserted state, the short side of the optical filter **160** becomes approximately parallel to an X direction (horizontal direction of the image pickup apparatus **100**), and is accommodated in the space between the opening **190** and the battery housing **170**.
[0049] In a case where the user intending to reinsert the optical filter **160** into the opening **190** presses the multifunction button **113** in the retracted state illustrated in FIG. **4**C to rotate the motor **201** in the direction opposite to that in the operation described above. Thereby, the optical filter **160** is inserted as illustrated in FIG. **4**A through the intermediate state illustrated in FIG. **4**B. In a case where the optical filter **160** moves to the position in the inserted state (first position) illustrated in FIG. **4**A, the optical filter driving circuit **137** stops the motor **201** by the detection signal of the position sensor (not illustrated), as in the retracted state.
[0050] This configuration can insert and retract the optical filter **160** to and from the optical axis **1000** without interfering with the internal parts or internal units of the image pickup apparatus **100**. Therefore, the optical filter **160** can be incorporated without making larger the image pickup apparatus **100**, and the optical filter **160** can be easily switched between the inserted state and the retracted state. In this embodiment, the configuration for switching the optical filter **160** between the inserted state and the saved state is not limited. For example, a mechanism for sliding the optical filter **160** in the horizontal direction of the image pickup apparatus **100** may be used to switch the optical filter **160** between the inserted state and the retracted state.
[0051] Referring now to FIGS. **6**A to **6**C, a description will be given of a control method of the image pickup apparatus **100**. FIGS. **6**A to **6**C explain optical paths of light incident on the image sensor **121**. FIG. **6**A illustrates the image pickup apparatus **100** in the inserted state, FIG. **6**B illustrates the image pickup apparatus **100** in the retracted state, and FIG. **6**C illustrates the optical paths in the sections of the inserted state and retracted state taken along lines B-B including the

optical axis **1000**. In the retracted state, the optical filter **160** does not exist in the section taken along the line B-B, but FIG. **6**C illustrates the optical filter **160** for comparing the optical path in the inserted state and the optical path in the retracted state.

[0052] In the inserted state of FIG. **6**A, the light incident on the image sensor **121** passes through the optical filter **160**, and the light passes through the optical path **862** indicated by a dashed line in FIG. **6**C and forms an image at an imaging point **863**. In the retracted state of FIG. **6**B, the light enters the image sensor **121** without passing through the optical filter **160**, but following the optical path **860** indicated by a solid line in FIG. **6**C, and forms an image at an imaging point **861**. Thus, the optical path length of the light incident on the image sensor **121** differs between the inserted state and the retracted state.

[0053] This embodiment sets a flange focal length such that the incident light forms an image on a plane (imaging plane) of the image sensor **121** in the retracted state. In the inserted state, the optical path length changes, and the focus position of the lens (in-focus position) changes to the rear side viewed from the light incident direction, which is a so-called rear focus state. Therefore, in the inserted state, the focus lens **141** may reach the driving limit and may not be able to be focused on a close object at the shortest object distance.

[0054] Referring now to FIG. **7**, a description will be given of a relationship between the depth of field and the imaging position. FIG. **7** illustrates a relationship between changes in the depth of field caused by driving the electromagnetic diaphragm **142** and the imaging points in the inserted and retracted states. Similar to FIG. **6**C, FIG. **7** illustrates light incident from the left side forming an image, and a solid line illustrates an optical path **860** in the retracted state and a broken line indicates an optical path **862** in the inserted state. A depth of field is an apparent in-focus range. The depth of field changes depending on the setting of the electromagnetic diaphragm **142**. In a case where an aperture in the electromagnetic diaphragm **142** is narrowed, the depth of field changes from a first depth of field **864** to a second depth of field **865**.

[0055] In a case where an in-focus state on a close object is not obtained in the inserted state, the depth of field is set to the first depth of field **864**, and an image is formed at the imaging point **863** in the inserted state outside the range of the first depth of field **864**. Therefore, by setting the depth of field to the second depth of field **865**, the imaging point **863** in the inserted state can fall within the range of the second depth of field **865** and the in-focus state can be obtained. Thus, narrowing the aperture in the electromagnetic diaphragm **142** to increase the depth of field can provide an in-focus state on a close object. As understood from the description with reference to FIG. **6**C, the in-focus state on the close object is obtained by transferring the inserted state to the retracted state.

[0056] In a case where an in-focus state on a close object is not obtained even in the inserted state, these methods are applied according to the imaging scene. For example, in a scene where there is a bright moving object within an imaging angle of view, the shutter speed is not changed, the aperture in the electromagnetic diaphragm **142** is narrowed (or made smaller), and the ISO speed is increased. An in-focus state can be obtained by narrowing the aperture in the electromagnetic diaphragm **142**, and a luminance change caused by the aperture change is adjusted by increasing the ISO speed.

[0057] In a scene with no bright moving object within an imaging angle of view, the aperture in the electromagnetic diaphragm **142** is narrowed and the shutter speed is lowered. An in-focus state can be obtained by narrowing the aperture in the electromagnetic diaphragm **142**, and a luminance change caused by the aperture change is adjusted by lowering the shutter speed.

[0058] In a scene where there is a dark moving object within an imaging angle of view, the shutter speed is not changed, the optical filter **160** is retracted, and the ISO speed is lowered. Retracting the optical filter **160** can provide an in-focus state, and lowering the ISO speed can adjust a luminance change in an object caused by retracting the optical filter **160**.

[0059] In a scene with no dark moving object within an imaging angle of view, the optical filter **160** is retracted and the shutter speed is increased. Retracting the optical filter **160** can provide an in-focus state, and increasing the shutter speed can adjust a luminance change caused by the aperture change.

[0060] The brightness/darkness of the object can be determined by the luminance signal read out of the image sensor **121**. Whether or not there is a moving object can be determined by the well-known object detection technology and image recognition technology. For example, any configurations may be used as long as the motion of the object is determined based on a change in motion vector between consecutive images. A moving object is, for example, an animal, a vehicle, or a waterfall, and it is determined that there is a moving object in a case where the object is moving at a certain speed or higher. In this embodiment, AF is performed by the image-plane phase-difference AF, but another AF method may be used.

[0061] Referring now to FIG. **8**, a description will be given of a method of determining whether or not an in-focus state on a close object can be obtained in the inserted state. FIG. **8** explains a defocus map (defocus map information). In AF that attempts to focus on a first object **871** for imaging at an imaging angle of view **870**, defocus amounts corresponding to distance information to the first object **871** and a second object **872** as the background within the imaging angle of view **870** are acquired. The defocus map includes defocus amount information within the imaging angle of view **870**. During AF, the defocus map is created by the MPU **130** based on the defocus amounts detected using the focus signals read out of the image sensor **121**.

[0062] Information on the thickness and refractive index of the optical filter **160** is previously set in the memory of the camera. The MPU **130** acquires information on the thickness and refractive index of the optical filter **160** and calculates the optical path lengths in the inserted state and retracted state. It is possible to determine based on the defocus map and the optical path length information that an in-focus state on the first object **871** as a target to be focused can be obtained in the retracted state, but an in-focus state on the first object **871** in the inserted state cannot be obtained. It is therefore possible to determine whether or not the current imaging faces a situation that can provide an in-focus state on a close object. Since the calculation of the optical path length in a direction along the optical axis **1000** depending on the presence or absence of the optical member is well-known, a detailed description thereof will be omitted.

[0063] Referring now to FIG. **9**, a description will be given of a control method of the image pickup apparatus **100** in a totally automatic imaging mode (auto mode). FIG. **9** is a flowchart of AF processing in the totally automatic imaging mode. As described above, the user can set a variety of imaging modes of the image pickup apparatus **100** by operating the mode switching dial **109**. The totally automatic imaging mode is an imaging mode in which the F-number, the shutter speed, and the ISO speed are automatically set according to the luminance (brightness) of the object.

[0064] In a case where the user starts imaging and AF processing starts, first, in step S**800**, the MPU **130** acquires the defocus map. Next, in step S**801**, the MPU **130** determines whether the optical filter **160** is inserted. In a case where the optical filter **160** is not inserted (or retracted), the flow proceeds to step S**812**. In step S**812**, the MPU **130** drives the focus lens **141** and ends this flow. On the other hand, in a case where the optical filter **160** is inserted, the flow proceeds to step S**802**. In step S**802**, the MPU **130** acquires optical filter information. The optical filter information is information about the thickness and refractive index of the optical filter **160** as described above.

[0065] Next, in step S**803**, the MPU **130** determines whether or not an in-focus state on the object is obtained in the retracted state but an in-focus state on the object is not obtained in the inserted state. The determination as to whether or not the in-focus state is obtained is made, for example, by calculating the defocus amount of the image based on a well-known method and by determining whether or not the calculated defocus amount falls within a predetermined range.

[0066] In a case where the in-focus state on the object is obtained both in the retracted state and in the inserted state, the flow proceeds to step S**812**. In step S**812**, the MPU **130** drives the focus lens **141** and ends this flow. On the other hand, in a case where the in-focus state on the object is obtained in the retracted state but the in-focus state on the object is not obtained in the inserted state, the flow proceeds to step S**804**. In step S**804**, MPU **130** displays a warning. This warning is issued, for example, by displaying an icon indicating a warning on a display unit such as the liquid crystal monitor **111** or the electronic viewfinder **112**. A written warning may be provided. This warning is for notifying the user of a change in imaging parameters.

[0067] Next, in step S**805**, the MPU **130** determines whether or not the luminance during imaging is higher than a threshold (luminance>threshold). As the unit of luminance, 1BV in the so-called APEX (additive system of photographic exposure) is set to one stage of luminance (photometric value). In a case where the luminance is higher than the threshold, the flow proceeds to step S**806**. On the other hand, in a case where the luminance is not higher than the threshold, the flow proceeds to step S**807**. Then, in step S**806** or step S**807**, the MPU **130** determines whether or not there is a moving object. In a case where it is determined in step S**806** that there is a moving object, the flow proceeds to step S**808**. On the other hand, in a case where it is determined in step S**806** that there is no moving object, the flow proceeds to step S**809**. In a case where it is determined in step S**807** that there is a moving object, the flow proceeds to step S**810**. On the other hand, in a case where it is determined in step S**807** that there is no moving object, the flow proceeds to step S**811**.

[0068] In step S**808**, the MPU **130** narrows the aperture in the electromagnetic diaphragm (lens diaphragm) **142** and increases the ISO speed. In step S**809**, the MPU **130** narrows the aperture in the electromagnetic diaphragm (lens diaphragm) **142** and lowers the shutter speed. In step S**810**, the MPU **130** retracts the optical filter **160** and decreases the ISO speed. In step S**811**, the MPU **130** retracts the optical filter **160** and increases the shutter speed. After any one of steps S**808**, S**809**, S**810**, and S**811** is performed, the flow proceeds to step S**812**, the MPU **130** drives the focus lens **141** and ends this flow. This control can provide an in-focus state on a close object according to an imaging scene.

[0069] Referring now to FIG. **10**, a description will now be given of a control method in a case where the inserted state and the retracted state are changed by the control in an F-number priority imaging mode. FIG. **10** is a flowchart of AF in the F-number priority imaging mode. In the F-number priority imaging mode, the F-number (aperture value) is set by the user. Therefore, the control flow does not change the F-number against the intention of the user.

[0070] After the user starts imaging and AF starts, first, in step S**820**, the MPU **130** acquires a defocus map. Next, in step S**821**, the MPU **130** determines whether or not the optical filter **160** is inserted. In a case where the optical filter **160** is not inserted (or retracted), the flow proceeds to step S**828**. In step S**828**, the MPU **130** drives the focus lens **141** and ends this flow. On the other hand, in a case where the optical filter **160** is inserted, the flow proceeds to step S**822**. In step S**822**, the MPU **130** acquires optical filter information. The optical filter information is information about the thickness and refractive index of the optical filter **160** as described above.

[0071] Next, in step S**823**, the MPU **130** determines whether or not the in-focus state on the object is obtained in the retracted state but the in-focus state on the object is not obtained in the inserted state. In a case where the in-focus state on the object is obtained both in the retracted state and in the inserted state, the flow proceeds to step S**828**. In step S**828**, the MPU **130** drives the focus lens **141** and ends this flow. On the other hand, in a case where the in-focus state on the object is obtained in the retracted state but the in-focus state on the object is not obtained in the inserted state, the flow proceeds to step S**824**. In step S**824**, MPU **130** displays a warning. This warning is issued, for example, by displaying an icon indicating a warning on a display unit such as the liquid crystal monitor **111** or the electronic viewfinder **112**. A written warning may be provided. This warning is for notifying the user of a change in imaging parameters.

[0072] Next, in step S**825**, the MPU **130** determines whether or not there is a moving object. In a case where it is determined in step S**825** that there is a moving object, the flow proceeds to step S**826**. On the other hand, in a case where it is determined in step S**825** that there is no moving object, the flow proceeds to step S**827**.

[0073] In step S**826**, the MPU **130** retracts the optical filter **160** and lowers the ISO speed. In step S**827**, the MPU **130** retracts the optical filter **160** and increases the shutter speed. After step S**826** or S**827** is performed, the flow proceeds to step S**828**, the MPU **130** drives the focus lens **141** and ends this flow. This control can provide an in-focus state on a close object according to an imaging scene.

[0074] Referring now to FIG. **11**, a description will be given of a control method in a case where the control changes the inserted state and the retracted state in a shutter speed priority imaging mode. FIG. **11** is a flowchart of AF in the shutter speed priority imaging mode. In the shutter speed priority imaging mode, the shutter speed is set by the user. Therefore, a control flow does not change the shutter speed against the intention of the user.

[0075] In a case where the user starts imaging and AF starts, first, in step S**830**, the MPU **130** acquires a defocus map. Next, in step S**831**, the MPU **130** determines whether or not the optical filter **160** is inserted. In a case where the optical filter **160** is not inserted (or retracted), the flow proceeds to step S**838**. In step S**838**, the MPU **130** drives the focus lens **141** and ends this flow. On the other hand, in a case where the optical filter **160** is inserted, the flow proceeds to step S**832**. In step S**832**, the MPU **130** acquires optical filter information. The optical filter information is information about the thickness and refractive index of the optical filter **160** as described above.

[0076] Next, in step S**833**, the MPU **130** determines whether or not the in-focus state on the object is obtained in the retracted state but the in-focus state on the object in the inserted state is not obtained. In a case where the in-focus state on the object is obtained both in the retracted state and in the inserted state, the flow proceeds to step S**838**. In step S**838**, the MPU **130** drives the focus lens **141** and ends this flow. On the other hand, in a case where the in-focus state on the object is obtained in the retracted state but the in-focus state on the object is not obtained in the inserted state, the flow proceeds to step S**834**. In step S**834**, MPU **130** displays a warning. This warning is issued, for example, by displaying an icon indicating a warning on a display unit such as the liquid crystal monitor **111** or the electronic viewfinder **112**. A written warning may be provided. This warning is for notifying the user of a change in imaging parameters.

[0077] Next, in step S**835**, the MPU **130** determines whether or not the luminance during imaging is higher than a threshold (luminance>threshold). In a case where the luminance is higher than the threshold, the flow proceeds to step S**836**. On the other hand, in a case where the luminance is not higher than the threshold, the flow proceeds to step S**837**.

[0078] In step S**836**, the MPU **130** narrows the aperture in the electromagnetic diaphragm (lens diaphragm) **142** and increases the ISO speed. In step S**837**, the MPU **130** retracts the optical filter **160** and lowers the ISO speed. After step S**836** or S**837** is performed, in step S**838**, the MPU **130** drives the focus lens **141** and ends this flow. This control can provide an in-focus state on a close object according to an imaging scene.

[0079] Referring now to FIG. **12**, a description will be given of a control method for maintaining the state selected by the user instead of automatically changing the inserted state and retracted state of the optical filter **160**. FIG. **12** is a flowchart of AF in a case where the optical filter **160** maintains the state selected by the user. In this embodiment, the insertion state and retraction state of the optical filter **160** can be set variable only by the selection of the user. In that case, the changing method from the inserted state to the retracted state cannot be applied in order to provide an in-focus state on a close object. Thus, this method narrows the aperture in the electromagnetic diaphragm **142** to deepen the depth of field, and thereby provides an in-focus state on a close object.

[0080] In a case where the user starts imaging and AF starts, first, in step S**840**, the MPU **130** acquires a defocus map. Next, in step S**841**, the MPU **130** determines whether or not the optical filter **160** is inserted. In a case where the optical filter **160** is not inserted (or retracted), the flow proceeds to step S**848**. In step S**848**, the MPU **130** drives the focus lens **141** and ends this flow. On the other hand, in a case where the optical filter **160** is inserted, the flow proceeds to step S**842**. In step S**842**, the MPU **130** acquires optical filter information. The optical filter information is information about the thickness and refractive index of the optical filter **160** as described above.

[0081] Next, in step S**843**, the MPU **130** determines whether or not the in-focus state on the object is obtained in the retracted state but the in-focus state on the object is not obtained in the inserted state. In a case where the in-focus state is obtained both in the retracted state and in the inserted state, the flow proceeds to step S**848**. In step S**848**, the MPU **130** drives the focus lens **141** and ends this flow. On the other hand, in a case where the in-focus state on the object is obtained in the retracted state but the in-focus state on the object is not obtained in the inserted state, the flow proceeds to step S**844**. In step S**844**, MPU **130** displays a warning. This warning is issued, for example, by displaying an icon indicating a warning on a display unit such as the liquid crystal monitor **111** or the electronic viewfinder **112**. A written warning may be provided. This warning is for notifying the user of a change in imaging parameters.

[0082] Next, in step S**845**, the MPU **130** determines whether or not there is a moving object. In a case where it is determined in step S**845** that there is a moving object, the flow proceeds to step S**846**. On the other hand, in a case where it is determined in step S**845** that there is no moving object, the flow proceeds to step S**847**.

[0083] In step S**846**, the MPU **130** narrows the aperture in the electromagnetic diaphragm (lens diaphragm) **142** and increases the ISO speed. In step S**847**, the MPU **130** narrows the aperture in the electromagnetic diaphragm (lens diaphragm) **142** and lowers the shutter speed. After step S**846** or S**847** is performed, the flow proceeds to step S**848**, the MPU **130** drives the focus lens **141** and ends this flow. This control can provide an in-focus state on a close object according to an imaging scene.

[0084] As described above, the image pickup apparatus **100** includes the optical filter **160** and the control unit (MPU **130**) configured to provide exposure control. The optical filter **160** is movable between the first position at which the optical filter **160** is inserted into the imaging range (opening **190**) of the image sensor and the second position at which the optical filter **160** is retracted from the imaging range. The first position is the position where the optical filter **160** covers the imaging range (the imaging area including the optical axis **1000**), and the second position is the position where the optical filter **160** does not overlap the imaging range. The control unit provides the exposure control in a case where the optical filter is located at the first position and the in-focus state cannot be obtained by moving a focus lens.

[0085] The control unit may provide the exposure control in a case where the optical filter **160** is located at the first position, the focus lens **141** is located at a driving limit position, and the in-focus state on a close object cannot be obtained. The control unit may provide the exposure control by changing the setting of the lens diaphragm (electromagnetic diaphragm **142**). The control unit may provide the exposure control by moving the optical filter **160** from the first position to the second position. The control unit may provide the exposure control when determining that the in-focus state on the object is obtained in a case where the optical filter is located at the second position, but the in-focus state is not obtained in a case where the optical filter is located at the first position, based on defocus map information and optical filter information.

OTHER EMBODIMENTS

[0086] Embodiment(s) of the disclosure can also be realized by a computer of a system or apparatus that reads out and executes computer-executable instructions (e.g., one or more programs) recorded on a storage medium (which may also be referred to more fully as a ‘non-transitory computer-readable storage medium’) to perform the functions of one or more of the above-described embodiment(s) and/or that includes one or more circuits (e.g., application specific integrated circuit (ASIC)) for performing the functions of one or more of the above-described embodiment(s), and by a method performed by the computer of the system or apparatus by, for example, reading out and executing the computer-executable instructions from the storage medium to perform the functions of one or more of the above-described embodiment(s) and/or controlling the one or more circuits to perform the functions of one or more of the above-described embodiment(s). The computer may comprise one or more processors (e.g., central processing unit (CPU), micro processing unit (MPU)) and may include a network of separate computers or separate processors to read out and execute the computer-executable instructions. The computer-executable instructions may be provided to the computer, for example, from a network or the storage medium. The storage medium may include, for example, one or more of a hard disk, a random-access memory (RAM), a read-only memory (ROM), a storage of distributed computing systems, an optical disk (such as a compact disc (CD), digital versatile disc (DVD), or Blu-ray Disc (BD)™), a flash memory device, a memory card, and the like.

[0087] Each embodiment can provide an image pickup apparatus, a control method of the image pickup apparatus, and a storage medium, each of which can provide an in-focus state on an object in a case where an in-focus state cannot be obtained by moving a focus lens while an optical filter is used.

[0088] While the disclosure has been described with reference to exemplary embodiments, it is to be understood that the disclosure is not limited to the disclosed exemplary embodiments. The scope of the following claims is to be accorded the broadest interpretation so as to encompass all such modifications and equivalent structures and functions. For example, this embodiment has discussed the configuration for controlling the insertion/removal of the optical filter **160** when the user presses the multifunction button **113**, but this embodiment is not limited to this example. For example, in a case where the insertion and removal of the optical filter **160** can be adjusted as one of the parameters for the exposure control, the image pickup apparatus **100** may automatically insert or remove the optical filter **160** according to the luminance (brightness) of the object.

[0089] This application claims the benefit of Japanese Patent Application No. 2022-032364, filed on Mar. 3, 2022, which is hereby incorporated by reference herein in its entirety.

Claims

1. An image pickup apparatus comprising: an image sensor; and at least one processor, and a memory coupled to the at least one processor, the memory having instructions that, when executed by the processor, perform operations as a control unit configured to perform exposure control for the image sensor in a case where an optical filter is located at a first position at which the optical filter is inserted into an imaging range of the image sensor and in a case where the control unit determines, based on defocus map information, that an in-focus state is obtained in a state where the optical filter is not located at the first position but the in-focus state is not obtained in a state where the optical filter is located at the first position.

2. The image pickup apparatus according to claim 1, wherein the first position is a position where the optical filter covers the imaging range, and the state where the optical filter is not located at the first position is a state where the optical filter doesn't cover the imaging range.

3. The image pickup apparatus according to claim 1, wherein the control unit performs the exposure control in a case where the optical filter is located at the first position, the focus lens is located at a driving limit position, and the in-focus state on a close object cannot be obtained.

4. The image pickup apparatus according to claim 1, wherein the control unit performs the exposure control by changing a setting of an aperture stop.

5. The image pickup apparatus according to claim 1, wherein the control unit displays a warning on a display unit in a case where the optical filter is located at the first position and the in-focus state cannot be obtained.

6. The image pickup apparatus according to claim 1, wherein the control unit changes a method for the exposure control based on luminance of an object.

7. The image pickup apparatus according to claim 1, wherein the control unit changes a method for the exposure control based on a motion of the object.

8. The image pickup apparatus according to claim 1, wherein the control unit changes a method for the exposure control based on an imaging mode.

9. The image pickup apparatus according to claim 8, wherein the control unit changes a method for the exposure control based on luminance and a motion of the object in a case where the imaging mode is set to a totally automatic imaging mode.

10. The image pickup apparatus according to claim 8, wherein the control unit performs the exposure control by retracting the optical filter from the first position and by lowering an ISO speed in a case where the imaging mode is set to an F-number priority imaging mode.

11. The image pickup apparatus according to claim 8, wherein the control unit performs the exposure control by retracting the optical filter from the first position and by increasing a shutter speed in a case where the imaging mode is set to an F-number priority imaging mode.

12. The image pickup apparatus according to claim 8, wherein the control unit performs the exposure control by changing a setting of an aperture stop and by increasing an ISO speed in a case where the imaging mode is set to a shutter speed priority imaging mode.

13. The image pickup apparatus according to claim 8, wherein the control unit performs the exposure control by retracting the optical filter from the first position and by lowering an ISO speed in a case where the imaging mode is set to a shutter speed priority imaging mode.

14. The image pickup apparatus according to claim 8, wherein the control unit performs the exposure control by changing a setting of an aperture stop and by increasing an ISO speed in a case where the optical filter is set to maintain a state selected by a user.

15. The image pickup apparatus according to claim 8, wherein the control unit performs the exposure control by changing a setting of an aperture stop and by lowering a shutter speed in a case where the optical filter is set to maintain a state selected by a user.

16. The image pickup apparatus according to claim 1, wherein the optical filter is an ND filter, a PL filter, or a soft filter.

17. A method of controlling an image pickup apparatus that includes an image sensor, the method comprising: a first determination step of determining whether an optical filter is in a state where the optical filter located at a first position at which the optical filter is inserted into an imaging range of the image sensor or a state where the optical filter doesn't locate at the first position; a second determination step of determining whether or not an in-focus state can be obtained by moving a focus lens while the optical filter is located at the first position; and an exposure control step of performing exposure control in a case where the second determination step determines that the in-focus state cannot be obtained.

18. A non-transitory computer-readable storage medium storing a program that causes a computer to execute the control method according to claim 17.